📄 WEEK 5 PROGRESS REPORT – MINOR PROJECT

Project Title

- AI-Powered Student Assistance Chatbot for Department of Technical Education, Government of Rajasthan

Group ID

- G-1

Branch & Section

- AI & DS – B

Date Range

- 16/02/2026 to 21/02/2026

1️⃣ EXECUTIVE SUMMARY

Current Sprint

- Sprint 5

Sprint Goal

- Interaction Modeling – Sequence Diagram Development

Status

- On Track

Summary of Work Done

- During Sprint 5, the team focused on modeling the interaction between system components using a Sequence Diagram.

- The objective was to represent how different system elements such as:

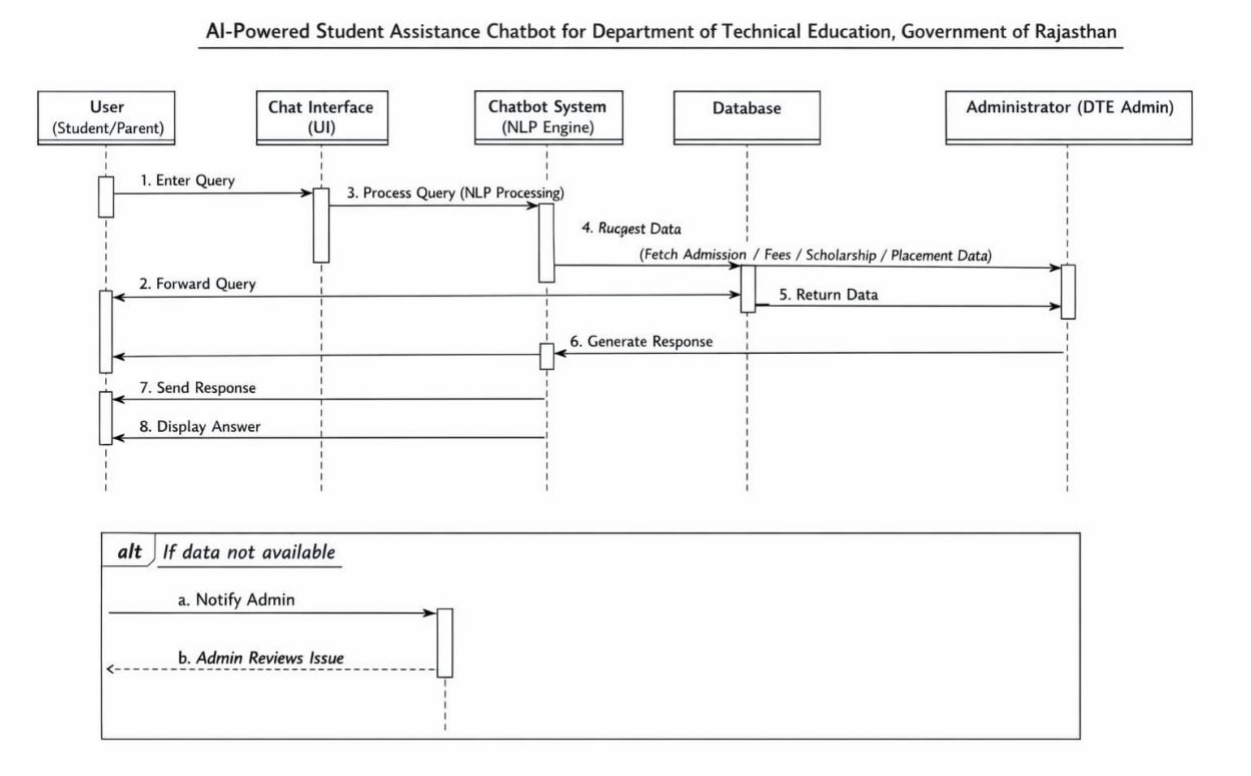
User (Student/Parent)

- Chat Interface (UI)
- Chatbot System (NLP Engine)
- Database
- Administrator
- interact step-by-step to process user queries.

The Sequence Diagram clearly shows the time-ordered flow of messages from query submission to response generation.

2 UML & OOAD PROGRESS

UML Diagram Completed This Sprint



The Sequence Diagram models the following interaction:

1. User enters query
2. Chat Interface forwards query to Chatbot System
3. Chatbot processes query using NLP
4. Chatbot sends request to Database
5. Database returns required data
6. Chatbot generates response
7. Response sent to Chat Interface
8. Answer displayed to user

Alternative Flow Modeled

- If data is not available:
- System notifies Administrator
- Admin reviews and updates data
- System resumes normal operation

Object-Oriented Concepts Applied

- Message Passing
- Interaction between objects
- Lifelines
- Activation bars
- Alternative (alt) fragment

3 TECHNICAL EXECUTION (DEVOPS & VERSION CONTROL)

Task Tracking

- Sequence modeling tasks created in Jira
- Interaction flow reviewed by team

Git Activity

- Sequence Diagram uploaded
- Sprint 5 documentation committed

Active Branches

- main

- interaction-modeling
- Merged Pull Requests
- Sprint 5 UML update merged successfully

CI/CD Status

- Not applicable (design modeling phase)

4 TESTING SUMMARY

- Logical Validation
- Verified correct message sequence
- Ensured no missing interaction steps
- Confirmed proper alternative flow handling

Issues Identified

- Need to handle multi-intent queries in future sprint
- Exception handling to be added during implementation phase

5 GOALS FOR NEXT SPRINT (SPRINT 6)

- Begin Backend Implementation
- Design REST APIs
- Connect Database with Application Layer
- Start Frontend Integration

6 SIGNATURES

Project Guide Signature: _____

Mentor Signature: _____